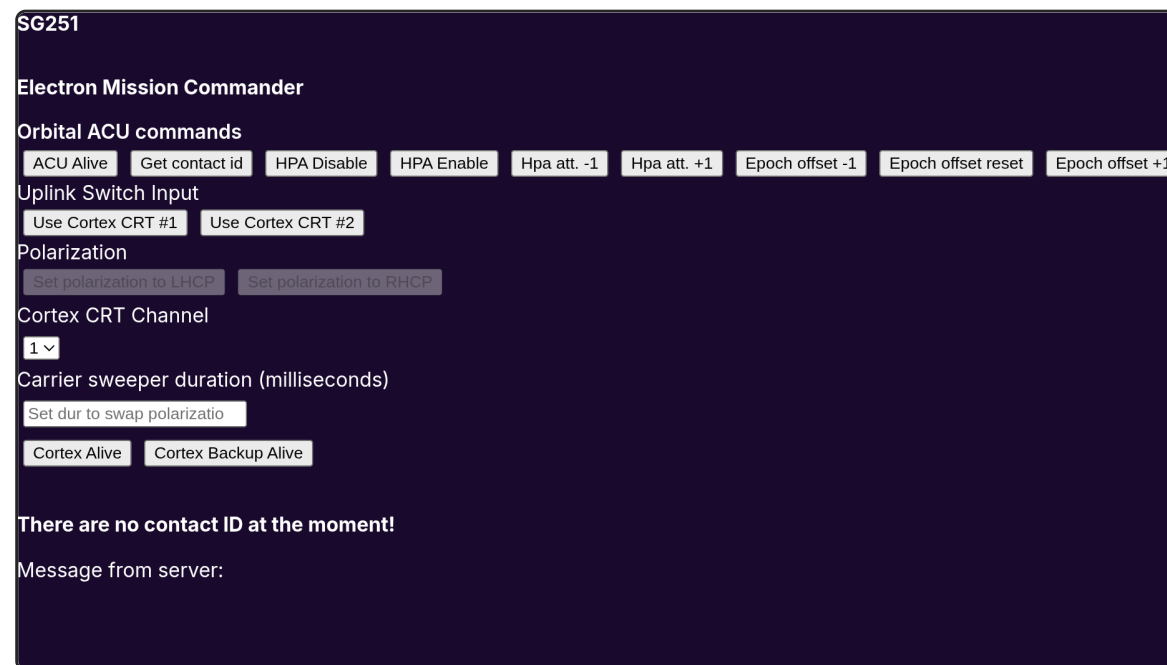


SpaceNintendo Control GUI

The SpaceNintendo GUI is a control GUI developed to provide operators within KSAT an interface to perform operations real-time operations for MAX Ground Stations running on the Electron + Spaceport M&C system.

The interface is designed with simplicity in mind, providing a unified layout that enables operators to perform a defined set of operations on the antenna system. A screenshot of the interface is shown below:



How to reach the SpaceNintendo GUI

Access to the GUI

The SpaceNintendo GUI is located on the Spaceport VM of the ground system. The GUI is controlled by RBAC, hence a user is required for TNOc and SGS to access the GUI. The SSH access can be configured in the inventory configuration of the system.

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Michael.Johansen@ksat.no

The RBAC configuration is performed as part of system deployment.

Connecting to the GUI

To access the interface, establish an SSH tunnel to the Spaceport associated with the antenna system you wish to control.

SSH tunnel can be configured in the command window:

```
ssh <your-AD-user-name>@<spaceport IP> -L 3000:localhost:8080
```

Here is an example to access the SG50 SpaceNintendo GUI:

```
ssh tnoc@10.203.127.195 -L 3000:localhost:8080
```

Here `tnoc` is the SSH user and `10.203.127.195` is the Spaceport IP for the SG50 antenna system.

After setting up the tunnel, you can access the Space Nintendo GUI on your local machine by opening your localhost in yur web browser: `http://localhost:3000` .

Space Nintendo Control Endpoints

The list of commands available in the GUI is described in the table below:

COMMANDS	DESCRIPTION	RESPONSES	SYSTEM
Maxar NextGen GS systems			
<code>ACU Alive</code>	Checks the ACU Status		Maxar NextGen
<code>Get contact id</code>	Outputs the Contact ID of the ongoing contact in the message field of the GUI		Maxar NextGen
<code>HPA Disable</code>	Disables the ACU HPA		Maxar NextGen
<code>HPA Enable</code>	Enables the ACU HPA		Maxar NextGen
<code>HPA att. -1</code>	Decreases attenuation with 1 dB		Maxar NextGen

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COMMANDS	DESCRIPTION	RESPONSES	SYSTEM
HPA att. +1	Increases attenuation with 1 dB		Maxar NextGen
Epoch offset -1	Offsets the antenna 1 second backwards in time		Maxar NextGen
Epoch offset reset	Reset Epoch Offset		Maxar NextGen
Epoch offset +1	Offsets the antenna 1 second forward in time.		Maxar NextGen
SG50, SG251, SG254			
Use Cortex CRT #1	Change the RF matrix input channel to CRT #1 for uplink	200 OK 500 Internal Server Error	SG50, SG251, SG254
Use Cortex CRT #2	Change the RF matrix input channel to CRT #1 for uplink	200 OK 500 Internal Server Error	SG50, SG251, SG254
Polarization	Switch between LHCP and RHCP polarization		SG50, SG251, SG254
Cortex CRT Channel	Define what Cortex CRT you want to perform sweep on.		SG50, SG251, SG254
Carrier sweeper duration (milliseconds)	Initiate a carrier sweep. Sweep duration configurable.	200 OK 422 Unprocessable Entity 500 Internal Server Error	SG50, SG251, SG254
Cortex Alive	Checks if the Safran Cortex CRT MW is alive	200 OK	SG50, SG251, SG254
Cortex Backup Alive	Check Status of backup Cortex sockets	200 OK	SG50, SG251, SG254

When performing an operation, a response message from the API will be produced under provided "The message from server" section of the GUI.

The responses from the operations performed is provided in standard HTTP format with a status message from the system:

`200 - Successful` - indicates that an HTTP request has succeeded.

`400 - Bad Request` - indicates that the server cannot or will not process the request due to something perceived as a client error. This means the problem lies with the client's request rather than the server itself.

`500 - Internal Server Error` - indicating that the server encountered an unexpected condition that prevented it from fulfilling the request.

Best Practices and Tips & Tricks

It is **not** recommended to keep the SSH tunnel running continuously. Once you have completed the required operations, close the connection to the Spaceport instance by typing exit in the command window. This will terminate the tunnel. The reason for this practice is to avoid leaving unnecessary open access routes into the ground station system.

If you need to open multiple instances of the GUI, for example if you want to do an operation on SG251 and SG254 simultaneously, the GUI's need to be opened in two separate GUI's on the browser. This can be done by chosing `port 3001` instead of `port 3000` for the second instance you want to open.

If you experience issue accessing the GUI, try to close and re-open the SSH tunnel before opening the localhost in your browser.

If the M&C system is redeployed on a system while the SSH tunnel was active, access to the GUI will be terminated. To open the GUI again, close the SSH tunnel and set up a new tunnel. As mentioned above, it is not recommended to keep the SSH tunnel open outside of contacts so as long you initiate the tunnel as part of the contact setup or during the contact, this is not an issue you will experience.

For questions or feedback regarding the GUI, don't hesitate to reach out to the Control Systems team in GN Products.